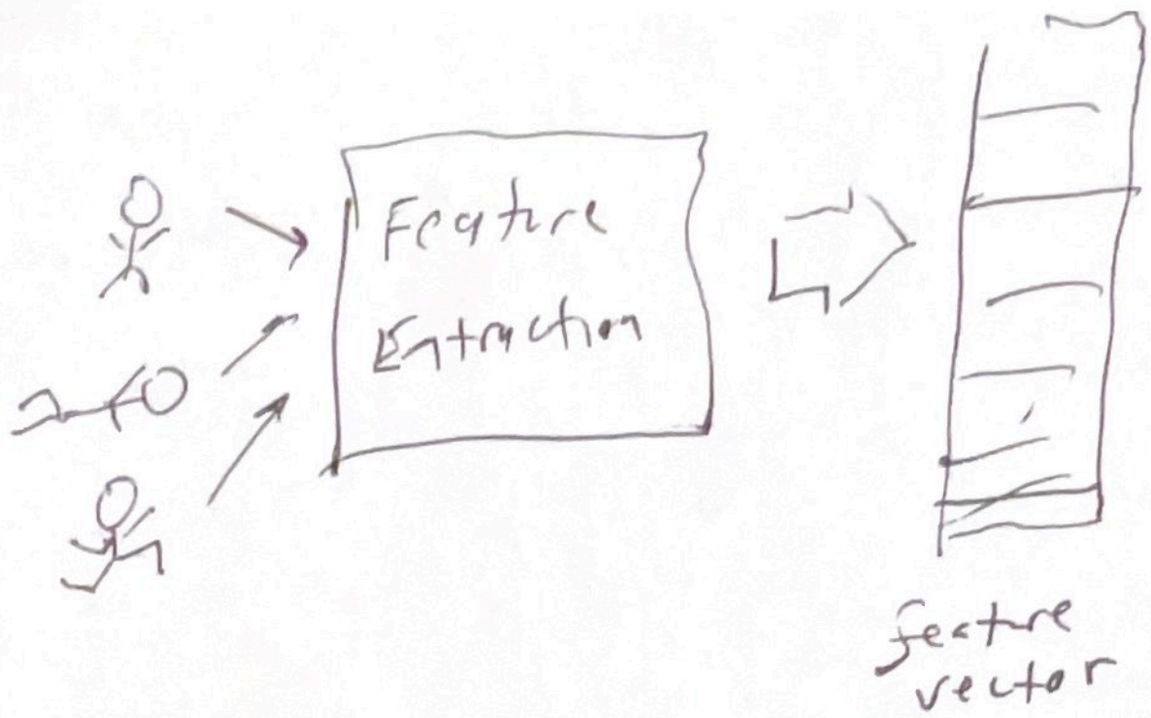
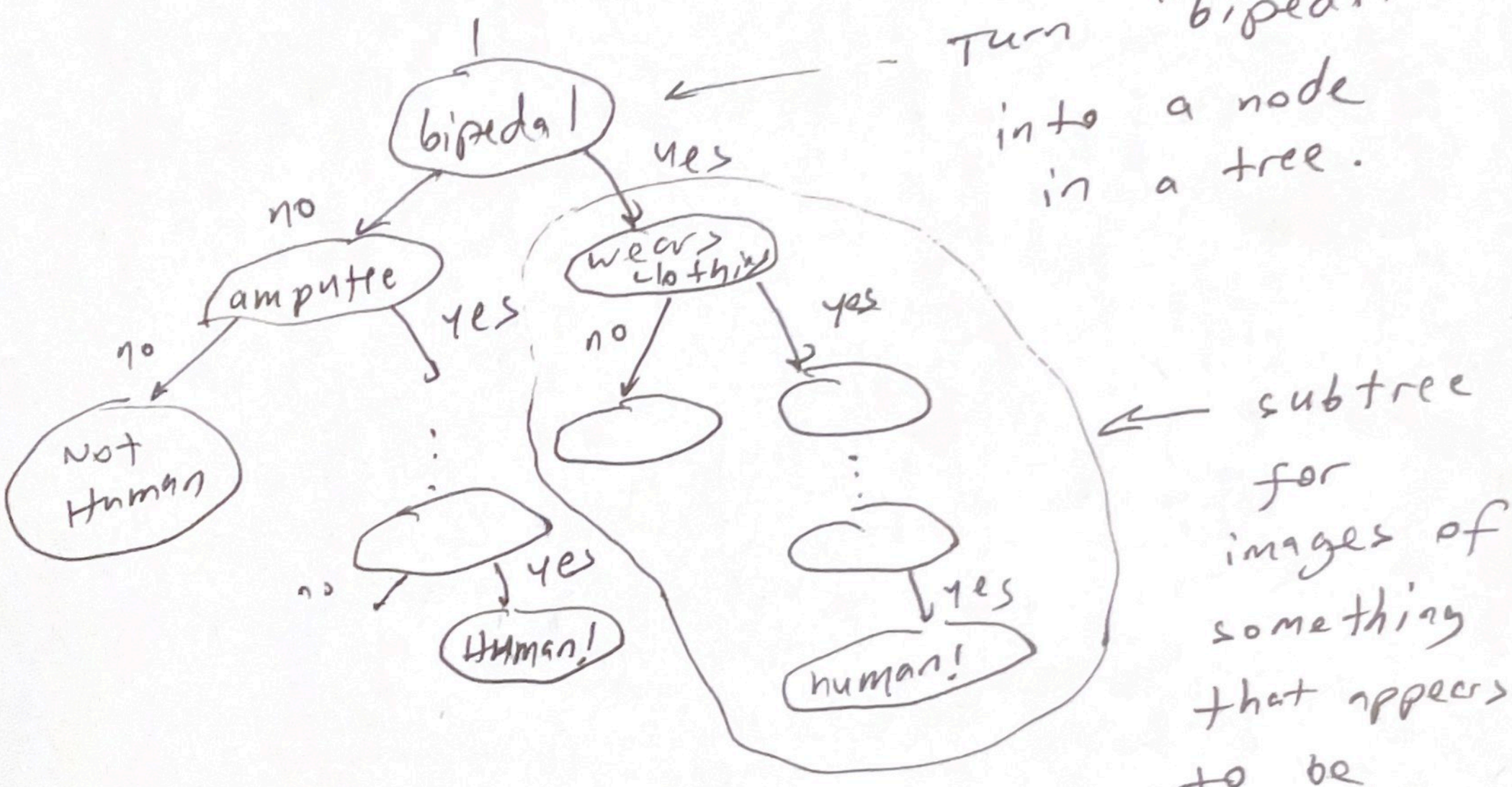
CSCI6032 lecture 17 Decision Trees



Now pick a feature that helps distinguish humans, e.g., bipedal.



Turn "bipedal" into a node in a tree.

← subtree for images of something that appears to be bipedal.

How do we pick the "best" feature to turn into a node?

One way is "information gain."

$$IG = H(P) - H(P|S)$$

← given S is true, this factoid provides some amount of information.

"Information gain" is a bit of a funny name, because "information gain" is really the reduction in Shannon entropy.

Feature selection (cont.)

If we pick the feature that maximizes information gain, this means we pick the feature that maximizes the reduction in entropy.

$$H(P) \geq H(P|S)$$

If S provides any usable information then

$$H(P) > H(P|S).$$

Information gain is thus always nonnegative.

Decision tree building algorithm

Let X denote the training set.

```
build_tree(X):
    if |X| < min_samples or reached max depth
        create leaf
        return leaf
    For each feature,
        if categorical, pick the split that maximizes IG.
        else numerical, pick the threshold that maximizes IG.
    S*, IG* = argmax_S IG(S | X)
        (S*: best split; IG*: information gain for best split;
         S: split; X: data. Note that X is shrinking with each recursion.)
    if IG* < min_gain:
        create leaf
        return leaf
    (L, R) = split the data X based on S*
    return Node(split=S*, left=build_tree(L), right=build_tree(R))
```

When we create a leaf, we put in it the posterior probabilities of the classes given all the decisions made up to that point.

id	Age	Smoker	Family Hx	Y
1	30	no	no	0
2	37	yes	no	0
3	41	no	yes	0
4	46	yes	no	1
5	50	no	yes	1
6	52	yes	yes	1
7	58	no	no	0
8	63	yes	no	1

Table 1: Input data X

let min_samples = 2
let max_depth = ∞
let min_gain = 0.01

root of decision tree

Y=0
Y=1

$$H(Y) = -\sum_{y \in \{0,1\}} P(Y=y) \log P(Y=y)$$

$$= -\left(\underbrace{\frac{1}{2} \log \frac{1}{2}}_{Y=0} + \underbrace{\frac{1}{2} \log \frac{1}{2}}_{Y=1} \right)$$

$$= -\left(-\tfrac{1}{2} - \tfrac{1}{2}\right) = 1$$

1) if $|X| <$ min_samples ?

False

2) For each feature:

Age? numeric.

Maximize $H(Y) - H(Y \mid age > th)$

Find threshold th.

S = age > 43.5

$$H(Y \mid S) = P[age \le 43.5]\, H_L + P[age > 43.5]\, H_R$$

H_L ← $\frac{3}{3}$ have $y=0$

$$H_L = -P[Y=0 \mid age \le 43.5] \log \underbrace{[P[Y=0 \mid age \le 43.5]}_{1})$$

$$- P[Y=1 \mid age \le 43.5] \log \underbrace{(P[Y=1 \mid age \le 43.5]}_{0})$$

$\frac{0}{3}$ have $y=1$

$$= 1 \cdot \log(1) + \underbrace{0 \cdot \log(0)}$$

$$\lim_{p \to 0} p \log p = \lim_{p \to 0} \frac{\log p}{1/p} = \lim_{p \to 0} \frac{1/p}{-1/p^2} = \lim_{p \to 0} (-p) = 0$$

$H_L = 0$

· 4 ones, 1 zero

$$H_R = \overbrace{-P[Y=0 \mid age > 43.5]}^{1/5} \log \overbrace{P[Y=0 \mid age > 43.5]}^{1/5}$$
$$\underbrace{- P[Y=1 \mid age > 43.5]}_{4/5} \log \underbrace{P[Y=1 \mid age > 43.5]}_{4/5}$$
$$= -0.2 \log_2 0.2 - 0.8 \log_2 0.8 \approx 0.722$$

$$H(Y|S) = \frac{3}{8} \cdot 0 + \frac{5}{8} \cdot 0.722 \approx 0.451$$

↑ S = "age > 43.5"

$$IG = H(Y) - H(Y \mid age > 43.5)$$

$$IG = 1 - 0.451 = 0.549 \text{ bits}$$

Next feature to consider is "smoker"
4 smokers & 4 non-smokers. $P[\text{smoker}] = 1/2$

$$H(Y \mid \text{smoker}) = \frac{1}{2} \cdot H_L + \frac{1}{2} \cdot H_R$$

$$H_L = -P[Y=0 \mid \text{not smoker}] \cdot \log P[Y=0 \mid \text{not smoker}]$$
$$- P[Y=1 \mid \text{not smoker}] \cdot \log P[Y=1 \mid \text{not smoker}]$$

3 out 4 non-smokers have $y=0$ 1 out of 4 has $y=1$

$$H_L = -0.75 \log_2 0.75 - 0.25 \log_2 0.25 \approx 0.811$$

$$H_R = -0.25 \log_2 0.25 - 0.75 \log_2 0.75 \approx 0.811$$

$$H(Y \mid \text{smoker}) = \frac{4}{8} \cdot 0.811 + \frac{4}{8} \cdot 0.811 = 0.811$$

$$IG(\text{smoker} \mid X) = 1 - 0.811 = 0.189 \text{ bits}$$

Now consider Family Hx.

3 yes, 5 no.

yes → ids {3, 5, 6}: labels (0, 1, 1) ⇒ $P(Y=1) = 2/3$

no → ids {1, 2, 4, 7, 8}: labels (0, 0, 1, 0, 1) ⇒ $P(Y=1) = 2/5$

$$H_{yes} = H(2/3) \approx 0.9183 \text{ bits}$$

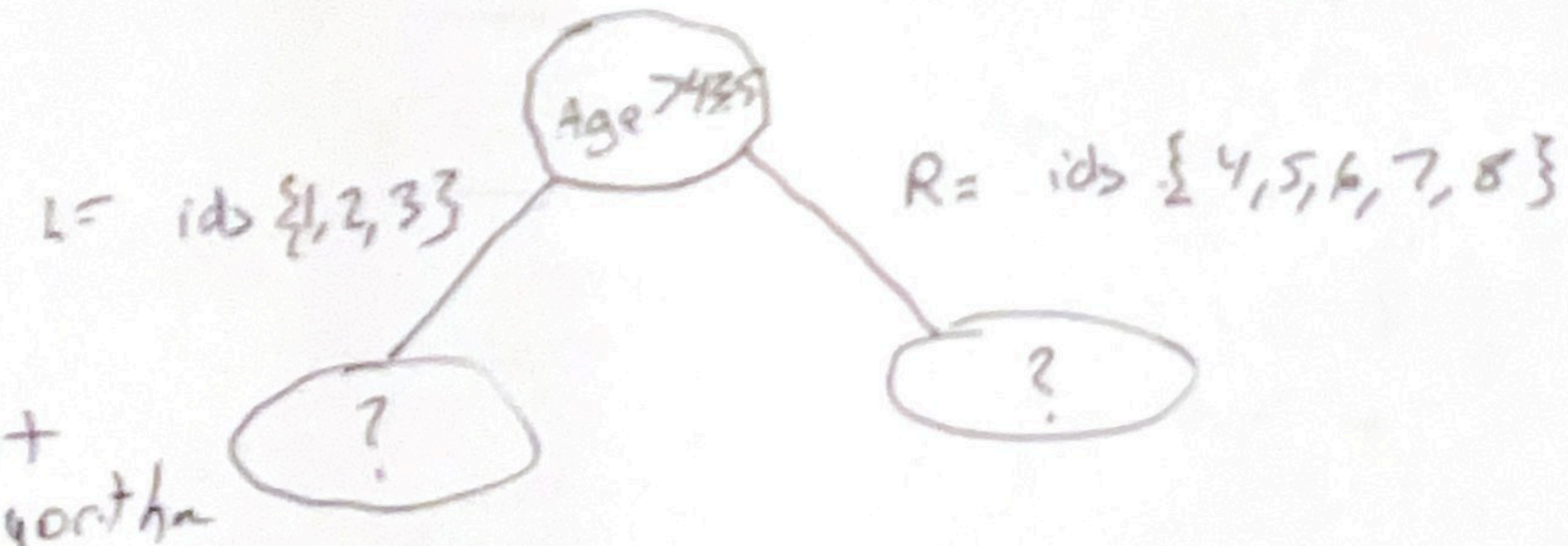
$$H_{no} = H(2/5) \approx 0.9710 \text{ bits}$$

$$H(Y \mid \text{Family Hx}) = \frac{3}{8} \cdot 0.9183 + \frac{5}{8} \cdot 0.9710 \approx 0.9512 \text{ bits}$$

$$IG(\text{Family Hx} \mid X) = 1 - 0.9512 \approx 0.0488 \text{ bits}$$

of the three features we have

	IG	
age > 43.5	0.549	← maximum IG
smoker	0.189	
Family Hx	0.0488	



Repeat the algorithm in each child

$X = \{1, 33\}$.

id	Age	smoker	Family Hx	Y
1	30	no	no	0
2	37	yes	no	0
3	41	no	yes	0

min_samples = 2 max_depth = ∞

build_tree(X):

1. "if |X| < min_samples" evaluates to false
   or reached max depth evaluates to false.

2. "For each feature:"
   - we reconsider age ...

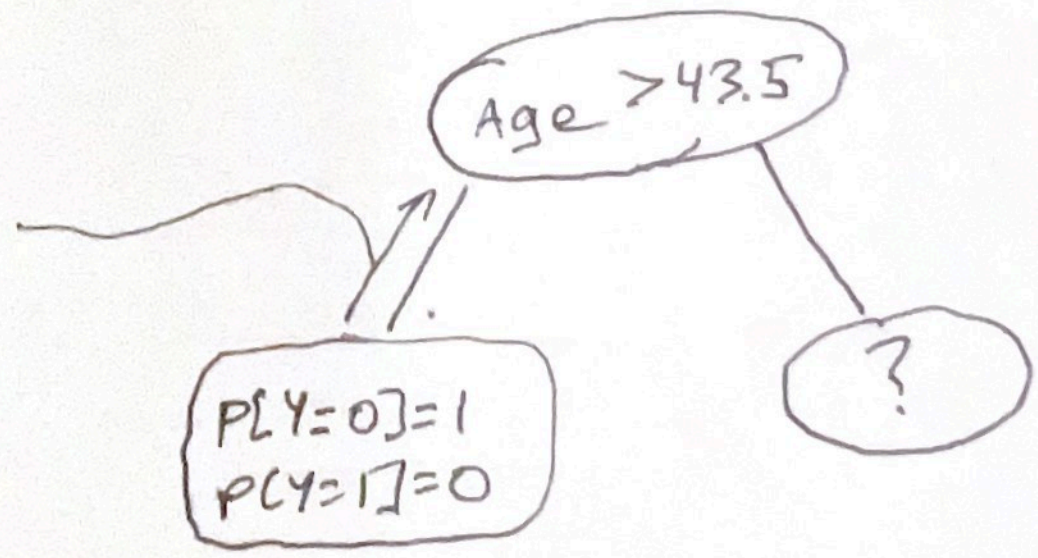
} we could mechanically measure the entropy of each feature, but $H(Y) = 0$ and all features $H(Y|s) = 0$.

3. $S^*, IG^* = \text{argmax}_S\ IG(S|X)$
   ↑
   0

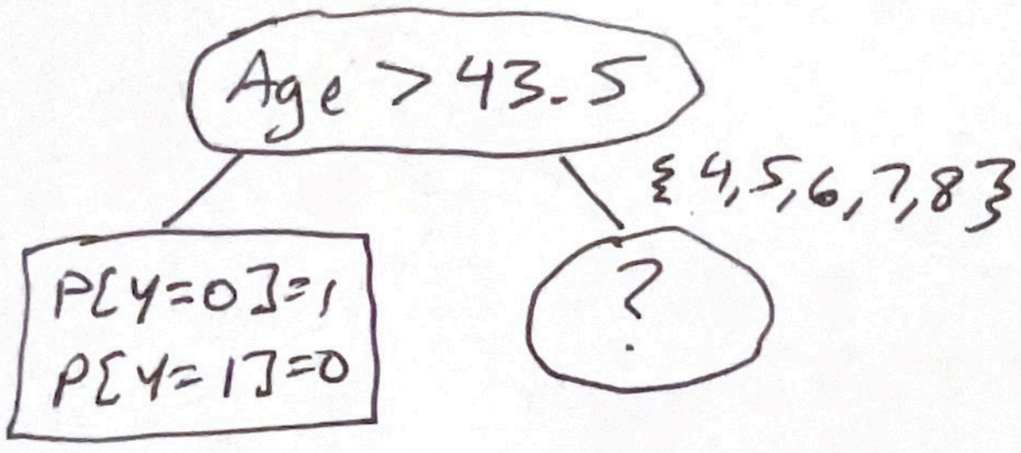
4. "if IG^* < min_gain:" evaluates to true. (min_gain = 0.01)
   ↑
   0

   create leaf ($P[y=0] = \frac{3}{3}$, $P[y=1] = \frac{0}{3}$)
   return leaf

Returned leaf becomes the left branch of the root.



build_tree, recursion back up unwinds and progresses to the right branch



id	Age	Smoker	Family Hx	Y
4	46	yes	no	1
5	50	no	yes	1
6	52	yes	yes	1
7	58	no	no	0
8	63	yes	no	1

For Age feature:

4 $y=1$ & 1 $y=0$

$$H(Y) = -(P[Y=0]\log P[Y=0] + P[y=1]\log P[y=1])$$

$$= -\left(\frac{1}{5}\log\frac{1}{5} + \frac{4}{5}\log\frac{4}{5}\right) \approx 0.722$$

$$H(Y|Age>55) = \frac{3}{5}H_L + \frac{2}{5}H_R$$

$$H_L = -(0+1\cdot 0) = 0, \quad H_R = -(0.5\log 0.5 + 0.5\log 0.5) = 1$$

$$H(Y|Age>55) = \frac{3}{5}H_L + \frac{2}{5}H_R = \frac{2}{5}$$

For smoker feature: 3 smokers, 2 non smokers

$$H(Y|smoker) = \frac{2}{5}\cdot H_{no} + \frac{3}{5}H_{yes}$$

$$H_{no} = -\left(\frac{1}{2}\log\frac{1}{2} + \frac{1}{2}\log\frac{1}{2}\right) = 1, \quad H_{yes} = -\frac{3}{3}\log\frac{3}{3} - \frac{0}{3}\log\frac{0}{3} = 0$$

$$H(Y|smoker) = \frac{2}{5}$$

For Family Hx feature: 3 no, 2 yes

$$H(Y|Family\ Hx) = \frac{3}{5}H_{no} + \frac{2}{5}H_{yes}$$

$$H_{no} = -\left(\frac{1}{3}\log\frac{1}{3} + \frac{2}{3}\log\frac{2}{3}\right), \quad H_{yes} = -\left(\frac{2}{2}\log\frac{2}{2} + \frac{0}{2}\log\frac{0}{2}\right) = 0$$

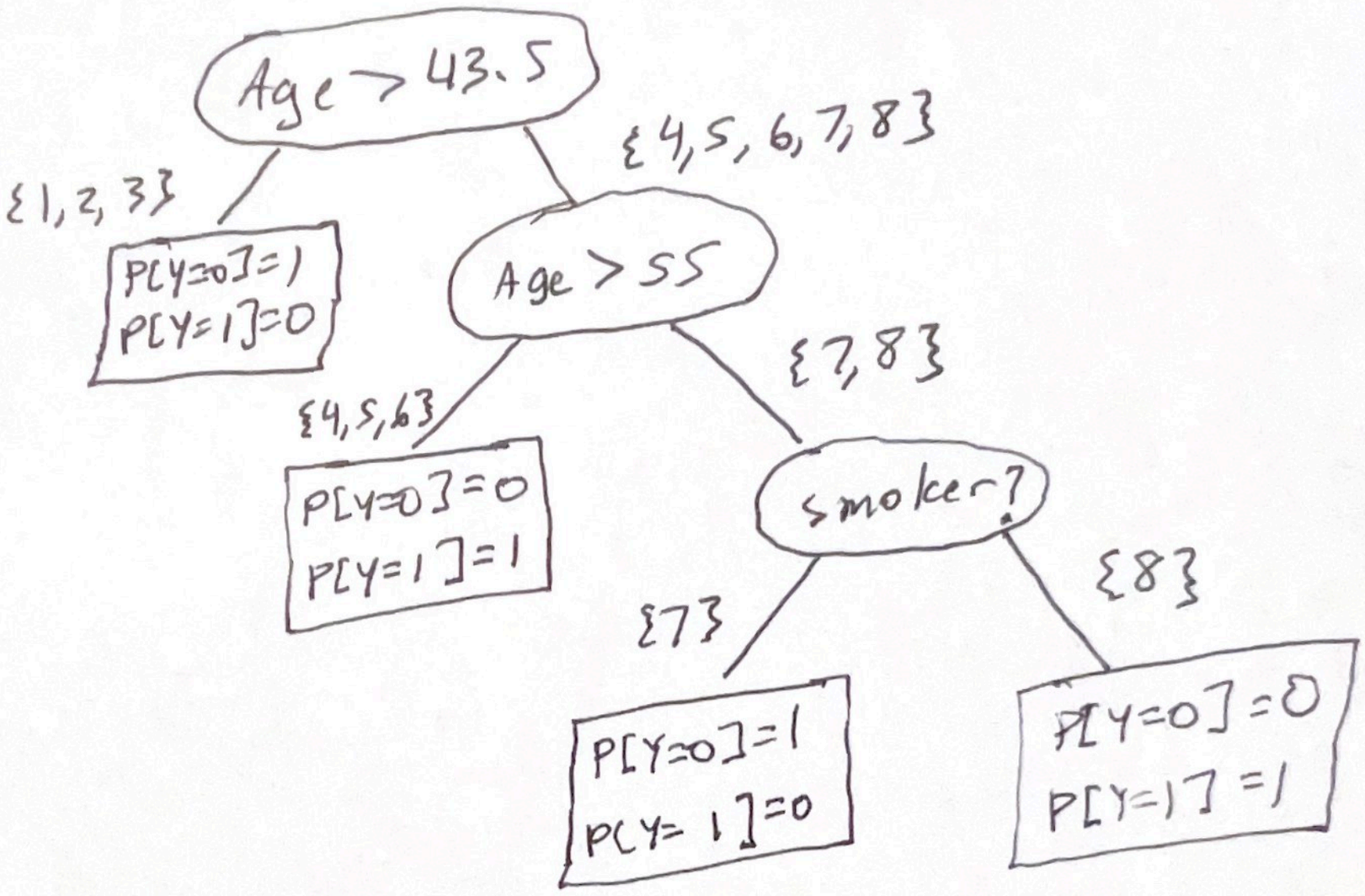
$$\approx 0.918$$

$$H(Y|Family\ Hx) \approx \frac{3}{5}\cdot 0.918 \approx 0.551$$

$$IG(Age>55|\underbrace{\{4,5,6,7,8\}}_{X}) = 0.722 - 0.4 = 0.322, \quad IG(smoker|X) = 0.722 - 0.4 = 0.322, \quad I(Y|Family\ Hx) \approx 0.171$$

tie

Because Age>55 and Smoker ~~are~~ have equal IG, we can pick either. Let's pick Age >55



In this example all leaf nodes have determinative posterior probabilities meaning they all achieve a probability of 1 for or the other class. In most cases this doesn't happen. In this case it occurs possibly because the sample size is so small.